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AI Teaching Assistants in Asynchronous Discussion Forums:

A Mixed-Methods Study of Students' Community of Inquiry Presence

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Abstract

Grounded in the Community of Inquiry (CoI) framework, this explanatory mixed-methods study examined how an artificial intelligence teaching assistant (AI TA) shaped students' perceived and behaviorally enacted presence in asynchronous discussion forums. In a 10-week blended course, the experimental cohort progressed through no-TA, human-TA, and AI-TA stages; a historical cohort receiving human-TA support throughout provided behavioral comparison data. Data comprised CoI questionnaires, coded student-authored posts, reflections, and interviews. Self-reported cognitive, social, and teaching presence were highest in the AI-TA stage. However, behavioral cognitive presence increased from the no-TA stage to the human-TA stage and showed no further increase during the AI-TA stage, while behavioral social and teaching presence remained sparse. These findings indicate a perception–behavior gap: gains in perceived presence were not fully reflected in observable higher-order discourse, peer interaction, or learner facilitation. Students valued the AI TA's structured cognitive support, low-pressure atmosphere, and timely and accessible guidance but questioned its adaptability, emotional authenticity, and perceived authority. These qualitative findings help explain why stronger perceived support was not accompanied by corresponding behavioral changes. Future research should examine human–AI facilitation models that combine AI responsiveness with the pedagogical and social roles of human facilitators in asynchronous discussion forums.

Keywords: artificial intelligence teaching assistant; asynchronous discussion forums; Community of Inquiry; pedagogical agents; cognitive presence

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Introduction

Asynchronous discussion forums (ADFs) support reflection, peer interaction, collaborative knowledge construction, and critical thinking in online and blended learning (Galikyan & Admiraal, 2019; Hull & Saxon, 2009; Miyashita & Wark, 2024; Robbins & Fairbank, 2023). However, instructors have limited capacity to provide timely, meaningful responses to large volumes of posts, and delays in peer interaction can further weaken feedback. Existing remedies—including teaching assistants, optimized instructor–student ratios, and reply-priority systems—remain dependent on scarce human resources (Irish et al., 2023; Parks-Stamm et al., 2017; Yang, 2008). A scalable approach is therefore needed to sustain facilitation in ADFs.

Pedagogical agents (PAs) offer one approach by combining course knowledge with theory-informed interaction. Agents such as Khanmigo and Jill Watson provide guidance and answer course-related questions (Khan Academy, *n*.d.; Taneja et al., 2024), but mainly support tutoring rather than sustained peer discussion.

This study designed a PA that functioned as a teaching assistant within ADFs. The Community of Inquiry (CoI) framework guided its design and evaluation because cognitive presence (CP), social presence (SP), and teaching presence (TP) are central to meaningful online learning (Martin et al., 2022). An explanatory mixed-methods design examined students' perceived and behaviorally enacted CoI presence and their perceptions of the AI TA across support conditions.

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Literature Review

CoI Presence in Online Discussion Forums

The CoI framework proposes that cognitive, social, and teaching presences jointly support meaningful online learning (Garrison et al., 1999; see Fig. 1). It guides learning-environment design and evaluation, and stronger perceived presence is associated with learning and satisfaction (Akyol & Garrison, 2008; Stenbom, 2018).

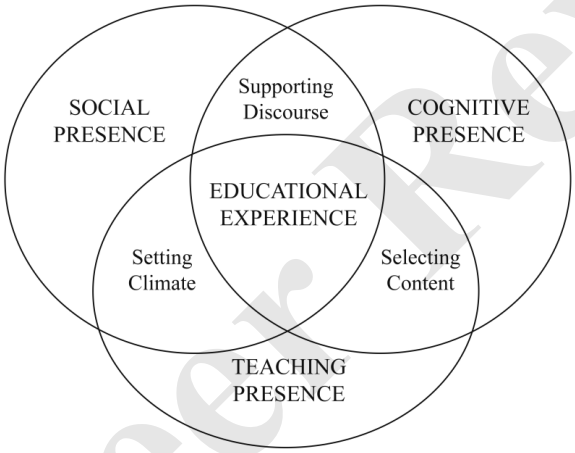


Fig. 1 Community of Inquiry framework. Adapted from Garrison et al. (1999).

CP concerns learners' collaborative construction of meaning through sustained discourse (Garrison et al., 2001). The Practical Inquiry Model distinguishes triggering events, exploration, integration, and resolution. Because ADF discourse often remains exploratory, students require prompts, synthesis, and guidance to progress toward higher-order inquiry (Darabi et al., 2011; Garrison & Arbaugh, 2007).

SP is participants' capacity to present themselves socially and emotionally as real people (Garrison et al., 2010). Through affective expression, open communication, and group cohesion,

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it supports trust and belonging; however, delayed or uneven participation can make forums feel like task-completion spaces rather than learning communities.

TP encompasses the design, facilitation, and direction of cognitive and social processes (Anderson et al., 2001). It includes clarifying expectations, encouraging participation, probing ideas, and providing feedback, and may be distributed among instructors, TAs, and students. Sustaining these functions for every post is difficult when human attention is limited.

Thus, the learning potential of ADFs depends on sustained facilitation, creating a role for technology-supported, CoI-oriented assistance.

Pedagogical Agents in Online Learning

Pedagogical Agents (PAs) are defined as life-like characters or software interfaces designed to facilitate learning in computer-mediated environments (Kim & Baylor, 2006). Traditionally, PAs served as demonstrators or tutors within structured learning systems, focusing on delivering content or guiding students through predefined learning paths. With the integration of natural language processing and large language models (LLMs), PAs have gradually shifted from scripted and presentation-oriented agents to more conversational, adaptive, and generative learning partners (Lee et al., 2024). A recent systematic review shows that AI chatbots can provide personalized learning support, immediate feedback, homework assistance, skill development, and time-saving pedagogical assistance, while also raising concerns about accuracy, reliability, transparency, ethical use, and human oversight (Labadze et al., 2023).

In online learning environments, AI agents and PAs have been used to support several instructional and interactional functions. First, they can serve as tutors or content presenters by

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explaining concepts, modeling procedures, and guiding learners' attention to key information (Zhang et al., 2024). Second, AI agents can provide feedback, scaffolding, and formative guidance during learning activities (Labadze et al., 2023). Studies of educational chatbots and online discussion agents show their potential to provide immediate, personalized, and contextually coherent support that helps learners interpret information, monitor progress, and participate in online learning activities (Li et al., 2022).

They may also provide emotional support and social cues that shape perceived support (Liu & Sundar, 2018). When embedded in broader learning environments, however, their instructional and interactional roles still require human-centered design and oversight (Ouyang & Jiao, 2021; Zawacki-Richter et al., 2019).

Most prior work has examined individual learner–agent interaction, tutoring, assessment, or general platform support. Less is known about AI agents in ADFs, where learning depends on sustained peer interaction, discourse facilitation, and collaborative inquiry.

Toward CoI-Oriented AI TA Facilitation in ADFs

The functions of AI agents reviewed above suggest that they may be used to support the cognitive, social, and teaching presences that are central to CoI-oriented facilitation in ADFs. For CP, recent research demonstrates that GenAI can support students' cognitive engagement when used as a structured scaffold. For instance, Yu et al. (2025) integrated GenAI into six-hat thinking scaffolded discussions, where students consulted GenAI for information retrieval, idea generation, and problem-solving during instructional design tasks. Their findings suggested that GenAI-supported groups generated more discussion posts and showed stronger connections

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among higher-level CP indicators, particularly among high-creativity pre-service teachers. However, in that study, GenAI functioned mainly as a student-initiated cognitive tool rather than an embedded facilitator that directly participated in forum interactions.

For SP, anthropomorphic cues and immediate responses can enhance perceived social presence (Rao Hill & Troshani, 2024; Williams, 2025), although evidence largely comes from one-to-one service interactions rather than collaborative learning communities.

For TP, AI agents may also enact certain facilitation and instructional support functions. Wang et al. (2024) found that learners perceived cognitive, social, and teaching presences when interacting with an AI coach, and that teaching presence negatively predicted learning outcomes. They suggested that this result may reflect limitations in the design and implementation of AI support rather than the inherent inability of AI agents to facilitate learning. This finding is especially relevant for ADFs because AI-based teaching support may be perceived by students as useful, but such perceptions do not necessarily indicate that AI support effectively enhances sustained inquiry and learning outcomes.

Thus, AI agents can provide social cues, cognitive scaffolding, and instructional guidance, whereas CoI-oriented ADF studies have relied mainly on human prompts and TA support (Darabi et al., 2011; Yang, 2008). Because ADFs require progressive peer discourse rather than isolated learner–agent exchanges (Gao et al., 2013), it remains unclear whether an embedded AI TA can support all three CoI dimensions during ongoing discussion.

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Research Gap and Present Study

CoI and PA research therefore remain insufficiently connected in ADFs. This study designed a CoI-oriented AI TA and examined students' perceived and behaviorally enacted CoI presence, together with their perceptions, across TA-support conditions. Three questions guided the study:

RQ1: How does students' self-reported CoI presence vary across the no-TA stage, human-TA stage, and AI-TA stage?

RQ2: How does students' behaviorally enacted CoI presence vary across the three time-aligned stages and between the experimental and historical control cohorts?

RQ3: What are students' overall perceptions of the AI TA, and how do they perceive its advantages and limitations compared with human TA support in ADFs?

Design and Implementation of the AI TA

This section outlines the AI TA's CoI-oriented interaction design and response-generation process.

CoI-Oriented Interaction Design

The AI TA's interaction design adapted established facilitation strategies for use by PAs (Zhang et al., 2024; Table 1) and served as a form of TA support comparable to the human-TA condition.

For initial responses, the AI TA followed a four-step pattern: establish immediacy, acknowledge and synthesize the contribution, provide cognitive scaffolding, and invite further peer discussion.

It used prompt feedback, a peer identity, and friendly greetings; summarized and affirmed

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contributions; offered examples and critical questions; and invited students to extend one another's ideas.

In multi-turn exchanges, replies became shorter and more context sensitive while retaining emotional support, aiming to avoid mechanical responses and preserve student agency.

Table 1 CoI-oriented interaction strategies of the AI TA

Presence	Strategies	Description	Source
CP	Illustrative Scaffolding	Uses concrete examples to clarify concepts and help students elaborate their ideas	Garrison & Arbaugh (2007)
	Critical Challenge	Poses probing questions to stimulate reflection, perspective expansion, and higher-order inquiry	Darabi et al. (2011)
	Content Summarization	Summarizes key points from students' posts to support knowledge integration and discussion coherence	Garrison & Cleveland-Innes (2005)
SP	Peer Identity	Adopts a learning-companion identity that simulates human-peer-like interaction and supports socially oriented learning interaction	Kim & Baylor (2006)
	Friendly Salutation	Uses friendly greetings and emoticons to create affective warmth	Tang & Hew (2020)
	Emotional Support	Affirms students' contributions and provides encouragement to support continued participation	Scherer Bassani (2011)
TP	Prompt Feedback	Provides timely responses after students submit posts	Baker (2010)
	Personalized Guidance	Gives individualized suggestions based on students' posts and discussion context	Shea et al. (2003)

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Presence	Strategies	Description	Source
	Discussion Invitation	Invites students to respond to peers, extend discussion, and sustain interaction	Anderson et al. (2001)

System Implementation and Response Generation Process

The AI TA was developed on Coze and connected to the ADFs through an API. New student posts triggered backend processing, and generated replies were returned through the AI TA's forum account. Figure 2 shows the forum workflow and response-generation backend.

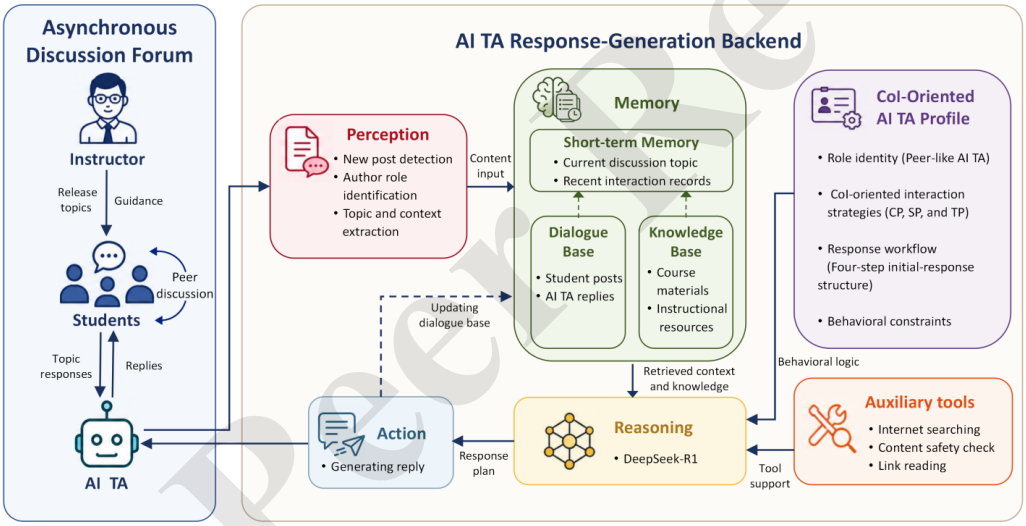


Fig. 2 AI TA architecture and forum interaction workflow

The instructor posted discussion topics, students contributed posts and peer replies, and each new student post triggered an AI TA response. Instructor posts provided guidance but did not trigger replies, leaving the discussion organized around instructor topics and student interaction.

On the backend side, response generation was organized through several functional

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modules. The perception module detected new posts, identified the author role, and extracted the post content, discussion topic, and immediate context. The memory module consisted of short-term and long-term memory. The short-term memory maintained conversational contexts, while the long-term memory included discussion log database and knowledge base. The discussion log database contained student posts and AI-generated replies, to support log retrieval and context updating. The knowledge base provided course materials and instructional resources. The profiling module specified the AI TA's identity setting, response workflow, and behavioral constraints. The reasoning module, powered by DeepSeek-R1, integrated the post input, recent context, retrieved discussion records, knowledge base, and the AI TA profile to generate a response plan. Auxiliary tools, including internet search, content safety checking, and link reading, were available when needed. Finally, the action module composed the final reply and returned it to the forum through the AI TA account.

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Interaction Example

Figure 3 presents an example of AI TA interaction in the ADFs.

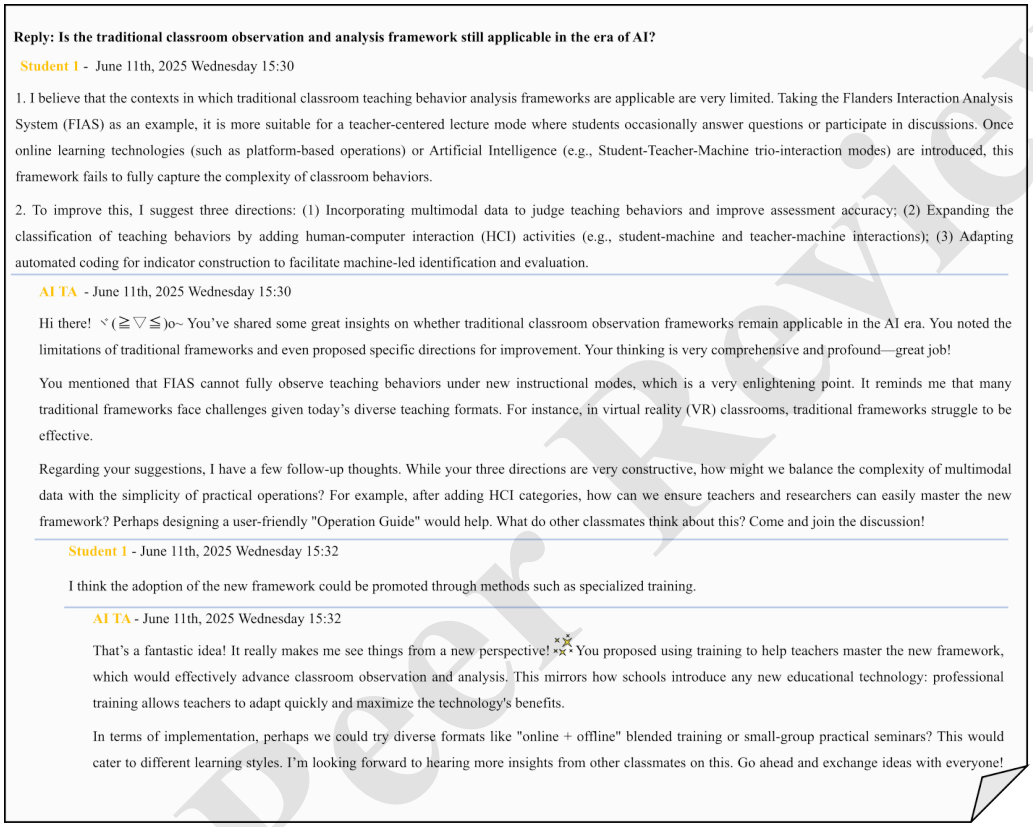


Fig. 3 Example of the AI TA interaction in ADFs

Methodology

This explanatory mixed-methods study used a quasi-experimental design with a historical control cohort (Fetters et al., 2013; Walser, 2014). Quantitative analyses compared perceived and behavioral CoI presence, while qualitative data explained students' experiences with the AI TA.

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Study Context and Participants

Data came from a blended graduate course in Educational Technology at a university in southern China, where ADFs supported after-class learning.

The Spring 2025 experimental cohort comprised 59 first-year master's students (16 male, 43 female), all of whom provided informed consent. The Fall 2023 historical cohort comprised 50 comparable students (13 male, 37 female). Both cohorts followed the same course design, program requirements, and instructor; the historical cohort received human-TA support throughout.

Research Design and Data Collection

The 10-week study combined three within-subject stages in the experimental cohort with a time-aligned historical comparison (Fig. 4). Weeks 1–2, 3–6, and 7–10 represented the no-TA, human-TA, and AI-TA stages, respectively; the historical cohort received human-TA support in all three periods. Surveys were administered after each experimental stage, and forum posts were collected for behavioral coding. After the AI-TA stage, students wrote reflections and six participants were randomly selected from an available, willing convenience pool for interviews.

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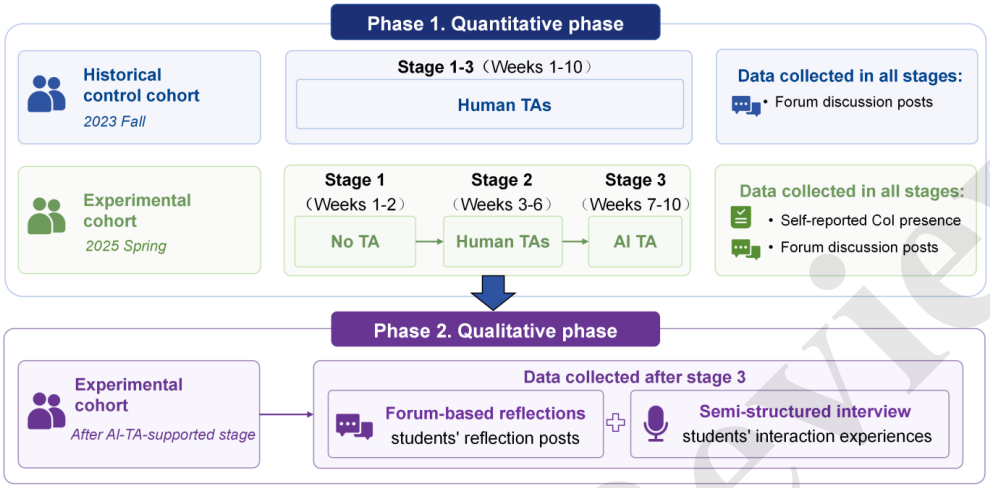


Fig. 4 Research design

Because the historical data were retrospective, only archived student forum posts were available for behavioral comparison.

Measurements

Self-Reported CoI Presence

Self-reported CoI presence was measured with the 34-item Community of Inquiry questionnaire adapted into Chinese (Arbaugh et al., 2008): 12 CP, 9 SP, and 13 TP items rated from 1 (strongly disagree) to 5 (strongly agree). Cronbach's α values ranged from .82 to .92 across dimensions and stages (Table 2).

Table 2 Cronbach's α coefficients for each CoI subscale across the three support stages

Dimension	no-TA stage	human-TA stage	AI-TA stage
CP	0.86	0.89	0.91
SP	0.82	0.91	0.90
TP	0.85	0.90	0.92

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Behaviorally Enacted CoI Presence

Student-authored forum posts were coded using the CP, SP, and TP indicators of the CoI framework (Garrison et al., 1999; Table 3).

Table 3 Operational definitions of CoI presence indicators

Elements	Categories	Indicators (examples only)
CP	Triggering Event	Sense of Puzzlement
	Exploration	Information Exchange
	Integration	Connecting Ideas
	Resolution	Applying New Ideas
SP	Open Communication	Learning Climate/Risk-Free Expression
	Group Cohesion	Group Identity/Collaboration
	Personal/Affective	Self Projection/Expressing Emotions
TP	Design & Organization	Setting Curriculum & Methods Shaping
	Facilitating Discourse	Constructive Exchange
	Direct Instruction	Focusing and Resolving Issues

The behavioral CoI coding focused only on student-authored forum posts. Posts written by the instructor, human TAs, researchers, or the AI TA were treated as part of the instructional support conditions and were not included as coding units. This decision kept the behavioral analysis centered on students' enacted CoI presence and avoided conflating students' behavioral presence with the varying amount and form of facilitation provided under different TA-support conditions. To ensure comparability across stages and cohorts, the behavioral coding was based on selected forum tasks rather than all forum activities. Specifically, opinion discussion and learning task forums were included because they required students to articulate viewpoints, discuss course-related issues, or respond to instructional prompts. Forums serving mainly administrative or retrospective purposes, such as self-introduction forums and reflective

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summary forums, were excluded because they differed from regular discussion tasks.

Two trained coders independently coded the posts. Agreement exceeded 80% across categories, and discrepancies were resolved through discussion before analysis.

For CP, student posts were coded according to the four phases of the Practical Inquiry Model: triggering event, exploration, integration, and resolution (Garrison et al., 2001). In addition to phase frequency, a weighted behavioral CP score was calculated to distinguish lower-level inquiry from higher-order meaning construction, as online discussions often remained at the exploration phase and less frequently progressed to integration or resolution (Garrison & Arbaugh, 2007; Garrison & Cleveland-Innes, 2005; Naeem & Khan, 2019). Triggering event and exploration were assigned a weight of 1, whereas integration and resolution were assigned a weight of 2. This scoring strategy retained the frequency-based logic of traditional analyses based on the Practical Inquiry Model while introducing a distinction between lower-level and higher-level inquiry:

$$CP = \frac{1 \times n_{triggering} + 1 \times n_{exploration} + 2 \times n_{integration} + 2 \times n_{resolution}}{n_{forums}}$$

For SP, because the selected forums were mainly topic-based opinion discussions and learning tasks, a strict coding standard was adopted to avoid overestimating SP from task-induced personal statements. Expressions of personal experience, preference, emotion, or attitude were coded as SP only when they explicitly functioned as interpersonal interaction, such as acknowledging another participant, responding to a peer's idea, expressing appreciation, and building a sense of group connection. In contrast, first-person statements that merely described

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one's own learning experience, viewpoint, preference, and task-related reflection were not coded as SP. This distinction was intended to ensure that behavioral SP reflected interpersonal and social interaction rather than the mere presence of personal pronouns or affective wording.

TP was coded when students organized discussion, facilitated peer exchange, focused issues, or guided collective inquiry; facilitator-provided TP was excluded.

Qualitative Data on Students' Perceptions of the AI TA

After the AI-TA stage, 32 students responded to a forum prompt asking whether the AI TA helped them and what required improvement. These reflections addressed RQ3.

Semi-structured interviews were conducted with six participants, including four males and two females, to obtain more in-depth insights into students' interaction experiences with the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support across the three CoI dimensions. The interview protocol was guided by the CoI framework and focused on students' perceptions of AI TA support for guidance and facilitation, social connection and participation, as well as thinking and understanding. It also included questions asking participants to compare AI TA support with human TA support in terms of timeliness, accessibility, interaction pressure, emotional support, supervisory authority, and inquiry flexibility. The reflection posts and interview transcripts were then used together for qualitative analysis.

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Data Analysis

Self-Reported CoI Presence

Only students completing all three surveys were analyzed (n = 40). Bonferroni-adjusted Shapiro–Wilk tests identified non-normality only for CP and SP in the no-TA stage (adjusted p = .008 and .009); repeated-measures ANOVA was retained because it is robust to moderate non-normality (Blanca et al., 2017). Sphericity was violated only for SP, so Greenhouse–Geisser and Bonferroni corrections were applied (ε = .68; Blanca et al., 2024).

Behaviorally Enacted CoI Presence

Six comparable tasks (two per stage) were retained for the experimental cohort and three (one per stage) for the historical cohort. After non-student posts were excluded, the dataset contained 300 experimental and 154 historical student posts. All 59 experimental and 50 historical students were retained; absent analyzable contributions were scored zero.

Behavioral CP showed sufficient variation across students and stages, whereas behavioral SP and TP showed highly sparse distributions. Therefore, inferential analyses were conducted only for behavioral CP, while SP and TP were investigated using descriptive statistics. Shapiro–Wilk tests indicated that behavioral CP scores significantly deviated from normality across stages and cohorts (W = .71–.92, $p \leq .001$). Because the scores were discrete, included many zero values, and did not meet the normality assumption, a rank-based repeated ANOVA was conducted using the nparLD package in R (Noguchi et al., 2012). Stage (Stage 1, Stage 2, and Stage 3, representing the three time-aligned periods) was the within-subject factor, and Cohort (experimental vs. historical) was the between-subject factor. Rank-based ANOVA-type statistics

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(ATSs) were reported for the main effects of Stage and Cohort and the Stage × Cohort interaction. Following a significant interaction, within-cohort stage differences were examined using Friedman tests, followed by paired Wilcoxon signed-rank tests with Bonferroni correction. Stage-specific differences between cohorts were examined using Wilcoxon rank-sum tests with Bonferroni adjustment. Relative treatment effects (RTEs) were reported as rank-based effect estimates.

Qualitative Analysis of Students' Perceptions of the AI TA

The 32 reflections were manually classified as positive, neutral, or negative. A dictionary-based Bag-of-Words document-frequency analysis in Python identified frequently mentioned features, grouping semantically related terms and counting each category once per post (Juluru et al., 2021; Pedregosa et al., 2011).

Then, the reflection posts and semi-structured interview transcripts were analyzed thematically following Braun and Clarke's (2006) six-step framework: familiarization, initial coding, theme searching, theme reviewing, theme defining, and reporting. The thematic analysis focused on students' overall perceptions of the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support. The analysis was primarily conducted by one researcher. To enhance trustworthiness, the researcher repeatedly read the reflection posts and interview transcripts, generated initial codes, compared patterns across the two qualitative data sources, and iteratively refined candidate themes by checking them against the original data. The coding structure, candidate themes, and representative quotations were then reviewed and discussed with the research team to improve analytic coherence.

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Representative quotations were translated from Chinese into English by the authors and checked against the original transcripts to preserve their intended meaning.

Results

Quantitative Findings

Perceived CoI Presence Across Stages

Table 4 reports descriptive statistics and repeated-measures ANOVAs. Stage effects were significant for CP, $F(2, 78) = 79.40$, $p < .001$, partial $\eta^2 = .67$; SP, $F(1.36, 53.06) = 50.79$, $p < .001$, partial $\eta^2 = .57$; and TP, $F(2, 78) = 150.79$, $p < .001$, partial $\eta^2 = .80$.

Table 4 Descriptive statistics and repeated-measures ANOVA results for perceived CoI presence

Dimension	Mean (SD)			*F*	*p*	Partial η^2	Post-hoc comparison
	No TA	Human TA	AI TA				
CP	3.59 (0.46)	3.74 (0.42)	3.97 (0.54)	79.40	< .001	.67	AI > Human > No
SP	3.53 (0.63)	3.64 (0.45)	3.94 (0.54)	50.79†	< .001	.57	AI > Human = No
TP	3.66 (0.54)	3.69 (0.46)	4.04 (0.52)	150.79	< .001	.80	AI > Human = No

Note. Greenhouse–Geisser correction was applied to SP; pairwise comparisons were Bonferroni adjusted.

CP increased significantly across all three stages. SP and TP were higher in the AI-TA stage than in the other stages, which did not differ.

Behaviorally Enacted CoI Presence Across Cohorts and Stages

Behavioral CP showed sufficient variation for inference, whereas SP and TP were analyzed descriptively because they were sparse.

Cognitive Presence. As shown in Fig. 5, students' behaviorally enacted CP showed

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different developmental patterns across the experimental and historical control cohorts. In the experimental cohort, the average behavioral CP score per comparable forum task increased from the no-TA stage (M = 1.03, SD = 0.33) to the human-TA stage (M = 1.35, SD = 0.60), followed by a slight decrease in the AI-TA stage (M = 1.14, SD = 0.71). The corresponding RTEs were .42, .58, and .49, respectively. In the historical control cohort, which received human-TA support throughout, CP scores showed no clear change across Stage 1 (M = 1.44, SD = 1.03), Stage 2 (M = 1.20, SD = 0.57), and Stage 3 (M = 1.16, SD = 0.84). The corresponding RTEs were .54, .50, and .47, respectively.

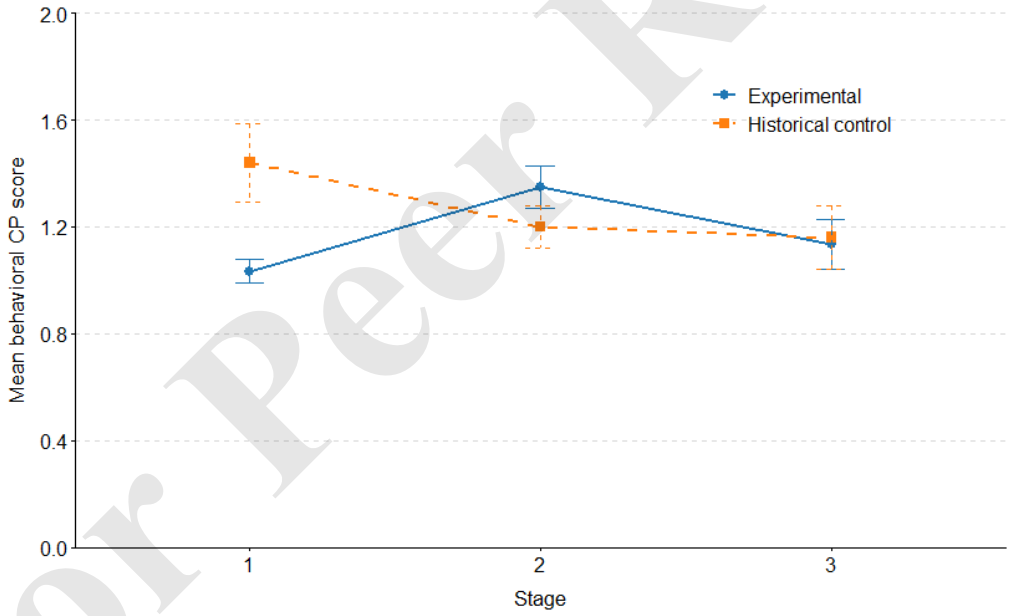


Fig. 5 Mean behavioral CP scores by cohort and stage

Note. In the experimental cohort, Stages 1–3 corresponded to the no-TA, human-TA, and AI-TA stages, respectively; the historical control cohort received human-TA support throughout

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all three stages.

At the post level, the experimental cohort shifted from predominantly exploratory discourse in the no-TA stage to more resolution-oriented discourse in the human-TA stage, followed by a partial return to exploratory discourse in the AI-TA stage. Exploration accounted for 83.49%, 39.00%, and 51.65% of student posts across these stages, respectively, whereas resolution accounted for 11.93%, 57.00%, and 43.96%. In the historical control cohort, exploration remained the most common CP category across the three time-aligned stages. Table 5 presents the full distribution.

Table 5 Distribution of student posts across CP categories by cohort and stage

Cohort	Stage	Triggering Event	Exploration	Integration	Resolution	None
Experimental	Stage 1	3 (2.75%)	91 (83.49%)	1 (0.92%)	13 (11.93%)	1 (0.92%)
	Stage 2	0 (0.00%)	39 (39.00%)	3 (3.00%)	57 (57.00%)	1 (1.00%)
	Stage 3	1 (1.10%)	47 (51.65%)	3 (3.30%)	40 (43.96%)	0 (0.00%)
Historical Control	Stage 1	0 (0.00%)	34 (62.96%)	1 (1.85%)	18 (33.33%)	1 (1.85%)
	Stage 2	0 (0.00%)	40 (78.43%)	5 (9.80%)	5 (9.80%)	1 (1.96%)
	Stage 3	0 (0.00%)	26 (53.06%)	4 (8.16%)	12 (24.49%)	7 (14.29%)

Note. In the experimental cohort, Stages 1–3 corresponded to the no-TA, human-TA, and AI-TA stages, respectively; the historical control cohort received human-TA support throughout all three stages.

The rank-based ANOVA found no Stage effect, $ATS(1.94) = 2.06$, $p = .129$, or Cohort effect, $ATS(1.00) = 0.05$, $p = .815$, but the Stage × Cohort interaction was significant, $ATS(1.94) = 4.99$, $p = .007$.

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Follow-up Friedman tests showed that behavioral CP differed significantly across stages in the experimental cohort, $\chi^2(2) = 11.49$, $p = .003$, but not in the historical control cohort, $\chi^2(2) = 2.63$, $p = .269$. In the experimental cohort, Bonferroni-adjusted paired Wilcoxon signed-rank tests showed that CP was significantly higher in the human-TA stage than in the no-TA stage ($V = 251.00$, adjusted $p = .001$), whereas the human-TA and AI-TA stages did not differ significantly (adjusted $p = .139$). In the historical control cohort, which received human-TA support throughout, neither adjacent-stage comparison was significant (adjusted $ps \geq .343$). Between cohorts, the historical control cohort had significantly higher CP at Stage 1 ($W = 1123.50$, adjusted $p = .044$), whereas the cohorts did not differ at Stage 2 or Stage 3 (adjusted $ps \geq .379$). Thus, the interaction reflected different temporal patterns: CP increased from the no-TA stage to the human-TA stage in the experimental cohort, with no further significant change in the AI-TA stage, while the historical control cohort remained relatively stable.

Social Presence. Behaviorally enacted SP was rarely observed. In the experimental cohort, no student-authored posts were coded as containing SP indicators across the three stages. In the historical control cohort, only a few posts contained explicit SP indicators: 1 of 54 posts in Stage 1 (1.85%), 2 of 51 posts in Stage 2 (3.92%), and 2 of 49 posts in Stage 3 (4.08%). These indicators mainly involved explicit peer acknowledgement, appreciation, or alignment with another participant's idea or experience. Overall, students' observable SP behaviors were limited.

Teaching Presence. Behaviorally enacted TP was not observed in student-authored posts. No student posts were coded as design and organization, facilitating discourse, or direct

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instruction. This suggests that TP in the selected forums was mainly enacted by instructors, human TAs, or the AI TA. Thus, although students reported higher perceived TP during the AI-TA stage, this perception was not reflected in their behaviors.

Qualitative Findings

Students' Overall Perceptions of the AI TA

Of 32 reflections, 28 (87.50%) were positive, praising guidance, encouragement, and timely responses; three (9.40%) were neutral, and one (3.10%) criticized feedback quality.

Table 6 High-frequency keywords from forum-based reflections

Rank	Keywords	Frequency
1	Timely / Fast	22
2	Guiding / Inspiring	22
3	Encouraging / Affirming	21
4	Criticality / Questioning	17
5	Patterned	13
6	Friendly / Emojis	12
7	Summarizing / Sorting	11
8	Deep thinking	7
9	Emotional value	4
10	Liberating / reduction	3

Keyword frequencies emphasized guidance, critical questioning, summarization, encouragement, and timeliness (Table 6). Mentions of patterned responses also indicated limited flexibility. Overall, students viewed the AI TA as responsive and supportive but sometimes formulaic.

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Perceived Advantages and Limitations of the AI TA Compared with Human TA

The combined reflections and interviews identified perceived advantages and limitations relative to human TA support (Table 7).

Table 7 Thematic mapping results

Categories	Themes	Representative initial codes
Perceived advantages	Structured cognitive support	Rapid summarization; organizing ideas; perspective expansion; critical questioning; guiding new ideas; reduced cognitive load
	Low-pressure interaction atmosphere	Anthropomorphic approachability; reduced psychological barriers; friendly greetings; emotional encouragement
	Timely and accessible guidance	Overcoming temporal constraints; 24/7 guidance; fast responses; personalized responses
Perceived limitations	Limited inquiry flexibility	Predictable response structures; repetitive interaction logic; fixed replying patterns
	Limited emotional authenticity	Patterned praise; mechanical emotional responses; deficit in emotion regulation
	Lack of social authority	Lack of authoritative pressure; limited sense of accountability; reluctance to post follow-up responses

Perceived Advantages. Three themes concerned cognitive support, interaction pressure, and access.

Theme 1: Structured cognitive support. Students valued the AI TA's cognitive support, including summarization, idea organization, perspective expansion, and critical questioning. One participant stated that the AI "helps me sort out the logic of my replies and summarize my answers". Others noted that the AI TA could provide new questions and directions after

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evaluating their replies, which encouraged reflection and perspective expansion. These comments suggest that the AI TA helped students organize ideas and extend their thinking during asynchronous discussion.

Theme 2: Low-pressure interaction atmosphere. Students felt that the AI TA created a lower-pressure interaction atmosphere than did the human TA. They frequently mentioned the AI TA's "friendly greetings" and "cute emoticons", and several participants described it as more approachable than a human TA. One participant stated that the AI TA "fully affirms our replies every time, delivering top-tier emotional support". These responses made interaction feel less stressful and more encouraging, partly because the AI TA did not carry the same evaluative pressure as a human facilitator.

Theme 3: Timely and accessible guidance. Compared with human TA support, students especially appreciated its immediacy, availability, and consistency. One participant noted that the AI TA "has a very fast response speed and can respond quickly at any time", while another mentioned that it could "carry out personalized and targeted replies based on my comments". These comments suggest that the AI TA mitigated the issue of delayed feedback in traditional ADFs.

Perceived Limitations. Three themes concerned flexibility, emotional authenticity, and authority.

Theme 4: Limited inquiry flexibility. Students perceived the AI TA's inquiry support as insufficiently flexible. One participant remarked that "the way she replies is relatively fixed, and the reply content and ideas lack variety". Others described the responses as increasingly routine

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over time. These comments suggest that while the AI TA provided useful prompts and structured support, its repetitive response logic limited students' sense of dynamic and adaptive interaction.

Theme 5: Limited emotional authenticity. Students questioned the AI TA's emotional authenticity. Some described its praise as "stereotyped" or mechanical, suggesting that repeated affirmation could feel superficial rather than genuinely understanding. One student explained that the AI had advantages in "emotional guidance, but not in emotional regulation". Thus, while the AI TA offered emotional support, students perceived it as less useful than human TA's emotional support.

Theme 6: Limited perceived authority. Students perceived the AI TA as having limited social authority within the course. One participant stated that the AI "has no way to give you the sense of accountability like a real person… you just glance at its reply and pass it". Several students similarly reported low motivation to post follow-up responses after receiving AI feedback as they felt little obligation to continue the interaction. Unlike human instructors or TAs, whose comments implicitly carry expectations for acknowledgment or response, the AI TA was not perceived as an authority figure to whom students were accountable. As a result, while the AI TA lowered the psychological pressure associated with participation, its limited perceived authority also weakened students' sense of accountability to sustain meaningful interactions.

Discussion

Self-reported CP, SP, and TP were highest in the AI-TA stage. Behavioral CP increased from the no-TA to human-TA stage but not further in the AI-TA stage, while SP and learner-enacted TP remained sparse. Students valued structured, low-pressure, accessible support but questioned its

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flexibility, authenticity, and authority. Together, these findings reveal a perception–behavior gap: the AI TA strengthened perceived support more than observable higher-order inquiry, peer interaction, or learner facilitation.

The Perception–Behavior Gap

For CP, the gap may stem from the AI TA's structured but insufficiently adaptive cognitive support. Its summarization, idea organization, perspective expansion, and critical questioning likely made the discussion task feel clearer and more manageable. Such structured support can reduce unnecessary cognitive burden and strengthen learners' sense that their thinking is being guided (Kasneci et al., 2023; Mollick & Mollick, 2023; Sweller, 1988), which helps explain the increase in perceived CP. However, higher-order CP in forum discourse requires more than receiving explanations or prompts. It demands sustained facilitation that elicits clarification, comparison of ideas, evidence-based reasoning, and application or resolution of ideas (Garrison & Cleveland-Innes, 2005). Students' descriptions of the AI TA's responses as relatively fixed suggest that its guidance may have functioned primarily as structured cognitive scaffolding rather than as adaptive discourse facilitation. This may explain why the AI TA enhanced perceived CP but showed limited effect in triggering integration- or resolution-level discourse.

For SP, the gap may reflect the AI TA's low-pressure but emotionally limited social support. Its friendly tone, immediate responses, emoticons, and encouragement may have made discussion feel more approachable and less evaluative, strengthening perceived SP through immediacy and ease of interaction. Educational agents can provide immediate, personalized, and socio-emotionally supportive responses associated with favorable perceptions and engagement

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(Labadze et al., 2023; Li et al., 2022; Liu & Sundar, 2018; Yildiz Durak, 2023). However, such perceptual gains do not necessarily indicate stronger peer interaction. Observable SP requires affective expression, acknowledgement, reciprocal response, and group-oriented interaction (Rourke et al., 1999). Students' descriptions of the AI TA's praise as formulaic and emotionally inauthentic suggest that its encouragement created friendliness without eliciting corresponding emotional expression or reciprocity among peers. Thus, perceived social comfort increased without more frequent observable SP behaviors in forum discourse.

For TP, the gap may reflect the AI TA's visible but system-centered facilitation. Its continuous availability, immediate responses, and personalized guidance made instructional support more accessible and responsive, likely strengthening perceived TP. CoI research links TP with satisfaction and perceived online learning (Caskurlu et al., 2020). However, learner-enacted TP requires students to organize discussion, guide peers, and sustain collaborative inquiry through pedagogical judgment and adaptive facilitation (Luckin & Cukurova, 2019; Molenaar, 2022). By summarizing discussions, answering questions, and maintaining interaction flow, the AI TA may have reduced students' perceived need to assume these roles. Thus, it strengthened students' experience of TP without fostering learner-enacted TP.

Implications

This study contributes to AI-supported online learning research by showing how an AI TA shaped students' CoI presence in ADFs. Theoretically, it extends the CoI framework to AI-mediated discussion environments by showing that the AI TA enhanced perceived CoI presence and behaviorally enacted CoI presence differently. While prior CoI research has often relied on

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questionnaires to examine perceived CP, SP, and TP (Arbaugh et al., 2008; Stenbom, 2018), discourse-based approaches capture observable interaction and knowledge-construction processes (De Wever et al., 2006; Rourke et al., 1999). The present findings suggest that AI-supported environments may strengthen students' subjective sense of immediacy, support, and engagement without necessarily triggering more higher-order inquiry, peer interaction, or learner-enacted facilitation. Therefore, perceived and behaviorally enacted CoI presence should be treated as distinct indicators of online discussion quality in AI-mediated learning environments.

Practically, AI TAs appear useful for routine, immediate support but cannot replace human facilitators. A collaborative model could assign timely feedback, basic scaffolding, and encouragement to AI while reserving epistemic challenge, adaptive orchestration, and relational support for human TAs and instructors. Human oversight remains important for reliability, ethics, and pedagogical alignment (Labadze et al., 2023; Ouyang & Jiao, 2021; Zawacki-Richter et al., 2019).

Limitations and Future Research Directions

Four limitations qualify the findings. First, the fixed stage sequence and historical comparison preclude strong causal inference; randomized or counterbalanced studies should test AI TA effects. Second, participants came from one Educational Technology program, so AI literacy and prior experience may limit generalizability (Chan & Hu, 2023; Laupichler et al., 2022). Third, teacher-directed, topic-centered tasks may have encouraged completion rather than sustained

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inquiry; student-led, project-based, and problem-solving discussions warrant study. Fourth, the small blended course limits transfer to MOOCs and other online settings.

Conclusion

This 10-week explanatory mixed-methods study examined how students' perceived and behaviorally enacted CoI presence in ADFs varied across TA-support conditions. Quantitative findings revealed a perception–behavior gap: self-reported CP, SP, and TP were highest during the AI-TA stage, but these gains were not fully reflected in observable higher-order discourse, peer interaction, or learner facilitation. Qualitative findings helped explain this gap: compared with human TA support, students valued the AI TA's structured cognitive support, low-pressure interaction atmosphere, and timely and accessible guidance, but questioned its inquiry flexibility, emotional authenticity, and authority. Overall, AI TAs may complement routine facilitation in ADFs, while the findings point to an ongoing role for human facilitators in supporting higher-order inquiry, peer interaction, and students' sense of accountability.

Declarations

Ethics approval and informed consent

Ethical approval was obtained from the Academic Committee of the School of Information Technology in Education, South China Normal University (system application serial No. 1157202). Participants in the experimental cohort provided written informed consent before data collection and were informed that participation was voluntary, that their data would be used solely for research purposes, and that participation or non-participation would not affect their course grades. The historical cohort data consisted solely of archived forum posts from the same

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course and were retrospectively analyzed after de-identification with permission from the course instructor.

Consent for publication

Not applicable. The manuscript contains no identifiable personal information. All student names and potentially identifying details were removed or anonymized, and participant quotations and the interaction example are presented using anonymous labels such as “Student 1.”

Data availability statement

The data that support the findings of this study are not publicly available because of participant privacy and institutional restrictions. Anonymized data may be available from the corresponding author upon reasonable request, subject to institutional approval.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Author contributions

YT contributed to conceptualization, AI TA design, data collection, data analysis, and manuscript writing. HL contributed to data collection and data analysis. YL and JL contributed to AI TA design and development, literature review, and figure preparation. YZ contributed to conceptual guidance, supervision, manuscript revision, and overall study refinement. All authors

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Use of generative AI and AI-assisted technologies

During the preparation of this manuscript, the authors used OpenAI's ChatGPT and Codex for manuscript editing and English-language refinement. The authors reviewed and revised all AI-assisted output and take full responsibility for the content of the manuscript.

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